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Divergence of investigative practice: The repudiation of Wundt

Wundt's limitations on introspection, experiment, and scientific psychology

Wundt's historical role was in some ways akin to the fate of the sorcerer's apprentice. He successfully mobilized some very effective practical techniques in the service of certain limited goals, but then found that these techniques turned into forces that had passed completely out of his control and were about to destroy the very framework within which he had put them to work. This was true of the link between introspection and experiment, which he advocated, and of the form of experimental psychology that he put into practice. Wundt, as has been observed, had many students but no genuine disciples. If he was the father of a discipline, he was a father whose fate is more reminiscent of the myth of *Totem and Taboo* than the myth of Chronos. Virtually everything that happened in modern psychology was a repudiation of Wundt, explicitly or implicitly. To understand this process, we must try to understand its starting point, namely, Wundt's conception of psychological practice.

Wundt's theory and practice of *introspection* diverged quite sharply from that of many of his students. On this issue (as on several others) he never emerged from the shadow of Kant, which meant that he basically accepted the object of knowledge to which the method of introspection corresponded but denigrated the method itself. He never doubted that the private consciousness was the object that psychology had to study, but he agreed with Kant and later critics like Comte and Lange that introspection was not the instrument that would transform this object into a scientific object. In fact, he went so far as to ridicule the introspectionist, likening him to a Baron Münchhausen attempting to pull himself out of the bog by his own pigtail.¹ Yet Wundt thought he had found a way to make consciousness available

to scientific inspection in spite of these objections. How did he propose to achieve this?

At the basis of Wundt's proposal there is a distinction, also made by Brentano at about the same time,² between actual introspection (*Selbstbeobachtung*) and internal perception (*innere Wahrnehmung*).³ It is a distinction between simply perceiving subjective events, being aware of them, and observing them in some methodical way. (As we saw in the previous chapter, some such distinction is in fact entailed by Kant's approach to the topic.) Now, as Wundt saw it, the problem for a scientific psychology of mind was the creation of conditions under which internal perception could be transformed into something like scientific observation. It was not enough to turn one's gaze inward in an attempt to give a systematic account of one's experience, for then one would not actually be observing ongoing events but more likely reflecting on what one thought one's experience had been. This was what the introspectionists of ill repute had done, and it was no basis on which to build a genuine science of mental life. The way out, for Wundt, was the manipulation of the conditions of internal perception so that they approximated the conditions of external perception. This manipulation was accomplished in the psychological experiment.⁴

In the laboratory, observation and report could follow immediately on the original perception, without time for self-conscious reflection. The conditions of internal perception would be closer to those for external perception where one reported on phenomena as they were happening. The conditions of psychological observation would thus approximate those of ordinary scientific observation. Second, according to Wundt, the laboratory made it possible to replicate specific experiences at will in order to observe them. We can make use of the fact that under certain circumstances identical external stimuli produce the same or very similar subjective experiences in order to scrutinize these experiences as often as we like and to be ready for them when they occur. This possibility of replicating observations again moves internal observation closer to normal scientific observation. The general idea was that internal perception could yield acceptable data for science only insofar as experimental conditions permitted a replication of inner experience at will. This rested on the assumption that there were external stimuli whose repeated presentation would reliably produce identical or near-identical perceptions. Wundt sought to achieve these conditions by limiting himself to relatively simple sensory stimuli and severely restricting the kinds of judgments required of experimental observers. In practice, so-called introspective reports from his laboratory were largely limited to judgments of size, intensity, and duration of physical stimuli, supplemented at times by judgments of their simultaneity and succession.⁵

A significant number of studies emanating from the Leipzig laboratory did not even contain this much "introspection." In these the experimental

data simply consisted of time measurements, as in the reaction-time studies to which Wundt attached much theoretical importance. Another set of studies used response measures that depend on the activity of the autonomic nervous system. A kind of introspection would sometimes be called upon to check on the effectiveness of experimental manipulations (e.g., fluctuations in the level of attention), but purely introspective data were not recognized as a basis for knowledge claims in the Leipzig laboratory. Wundt would have been horrified to find himself classified as an introspectionist. Nevertheless, he held fast to the traditional notion that the proper object of investigation for the field of psychology was the human mind. This tension between method and subject matter lies at the root of psychological research practice in its earliest period.

Wundt had his own set of methodological prescriptions for dealing with this tension. They led him to impose very severe limitations on the scope of the experimental method in psychology. To understand the early history of scientific practice in psychology, one has to appreciate the Janus-like face presented by its most important figure. The same Wundt whose laboratory functioned as the model and inspiration for numerous imitators was also the source of a mounting stream of restrictions on the use of the experimental method in psychology.

One set of limitations was that which affected the *range* of topics and problems to which the experimental method could be validly applied. This range was defined by the reliable coordination of specific external conditions and specific internal conditions open to internal perception. Effectively, this kind of coordination could only be achieved in experiments on sensation and perception. Wundt had hoped that a similar coordination might also be achieved on the response side, initially involving time measurements⁶ and later also measurements based on autonomic arousal. But essentially this hope proved to be unjustified, which may have contributed to his tendency to become less and less sanguine about the prospects for the experimental method as time went on. At all times Wundt explicitly excluded certain parts of psychology from the possibility of effective experimental investigation.⁷ Specifically, these were the processes of thought and all but the simplest affective processes. Here there seemed no hope of achieving that direct and reliable coordination of external conditions and internal perception that an effective psychological experiment presupposed.

In the course of his long career Wundt at times expressed himself with varying degrees of optimism or pessimism as to whether certain areas of psychology were likely to succumb to the experimental method. Such opinions, however, did not involve any change in basic principles. What he always accepts, as an inescapable corollary of his fundamental principles of methodology, is the notion that the area of psychology cannot be coextensive with the area of experimental psychology. The appropriateness of

experiment for the solution of various types of psychological problems is a matter of degree. At one extreme are problems for which the experimental method provides an excellent source of valid data, at the other extreme are problems that are quite unsuitable for experimental investigation. Wundt consistently assigns problems in the areas of sensation and perception to the top end of this quasi scale and problems in the areas of thinking, affect, voluntary activity, and social psychology to the bottom end. In between there are areas, such as memory, imagery, and attention, where the experimental method is partially appropriate. Precisely where the line is to be drawn at any time depends partly on technical developments and partly on the optimism of the investigator, and hence the line is historically variable. But such decisions do not affect the fundamental frame of reference within which they are made.

Even in his period of youthful enthusiasm for the experimental method Wundt recognized the need for another, nonexperimental type of psychology. This was *Völkerpsychologie*, an untranslatable term referring to a kind of social psychology based on the historical, ethnographic, and comparative analysis of human cultural products, especially language, myth, and custom.⁸ For all his interest in experimentation Wundt was sufficiently immersed in the German tradition of objectifying *Geist* to reject the notion that a study of the isolated individual mind could exhaust the subject matter of psychology. But as his conception of psychological experimentation made precisely the isolated individual consciousness its object of study, the only recourse was to supplement it by a different kind of investigation directed at mind in its objective manifestations. These investigations occupied Wundt increasingly in his later years and resulted in the series of volumes that he published under the general title *Völkerpsychologie* from 1900 onward. He now expressed himself to the effect that this was the more important part of psychology, which was destined to eclipse experimental psychology.⁹

Although Wundt's formulation of the precise relationship between experimental psychology and its complement underwent several changes in the course of his long life,¹⁰ he stuck to three fundamental points: First, that experimental psychology could never be more than a part of the science of psychology as a whole; second, that it needed to be supplemented by a branch of psychological studies that was devoted to the investigation of human mental processes in their social aspects; and third, that this latter type of study was able to make use of information that was no less objective than the data of experimental psychology.

But Wundt not only imposed severe limitations on the *range* of questions that the experimental part of psychology could handle, he also held a general theory of scientific method that tended to circumscribe the role of experimental evidence in the scientific enterprise as a whole. His view of experimental science was decidedly antiinductivist.¹¹ The aim of scientific

explanation was the demonstration of the logical coherence of the world by revealing its underlying causal relations. One performed experiments in pursuit of this aim, not to accumulate data from which generalizations might follow.

The world of physical science, according to Wundt, had been created by ignoring "the subjective elements of perception." This made possible the universe of physical causality but also implied that if the neglected elements were to be scientifically investigated there would emerge a science that would "supplement physics in the investigation of the total content of experience." This science was psychology.¹² Whereas physical science would frame its explanations in terms of physical causality, psychology, by analogy, would be concerned with something Wundt called *psychic causality*. The specific principles of psychic causality that Wundt attempted to formulate¹³ always remained a bit vague, but this does not affect the fundamental role this concept played in his approach to psychological investigation. Whatever their precise form, Wundt believed that there were psychological determinants at work in experience and that it was the task of psychological experiments to explore their mode of operation. Psychological investigation had to be experimental, rather than merely observational, because what it sought to illuminate were not static structures but directed causal processes.¹⁴ Wundt was, however, insistent that the laws of psychic causality would be qualitative in nature, and this limited the functionalism of his experimental approach.

It is clear that Wundt's practice involved a certain leap from the experimental to the theoretical level, and this is closely related to the tension between method and object of investigation, which has already been noted. For psychological experiments to be relevant to psychic causality, it was not necessary for their data to consist of introspective reports. It was necessary, however, that they be conducted with individuals who were in a position to provide such reports. For the aim of such investigations was the exploration of psychic causality, and this, according to Wundt, only operated within fully conscious human experience. If one worked with animals, or with individuals in pathological states of consciousness, one had given up psychology's proper object of investigation. Thus, although Wundt hardly qualifies as an introspectionist on the level of experimental practice, he always remained mentalistic on the level of theoretical explanation. Without a mentalistic object psychology would not have a distinct subject matter of its own, although without an experimental practice based on physical manipulation and measurement it would have nothing trustworthy to say about its subject matter.

This profound duality in Wundt's investigative practice was undoubtedly a reflection of his failure to cut himself off from his divergent physiological and philosophical roots. He hardly altered the physiological techniques he had inherited, and neither did he propose any fundamental change in the

object of investigation, which he had taken over from certain trends in philosophy. But he made no significant changes because it is abundantly clear he had no desire to make such changes. Wundt had no interest in two goals that were to become crucial for his successors and that were to lead to the transformation of modern psychology: He had no interest in the possible practical applications of psychology and he had no interest in converting psychology into an independent discipline without ties to philosophy.

Wundt belonged to a generation of German academics for whom the refusal of practical social involvement outside the university was effectively a condition of academic freedom.¹⁵ At the same time he was quite content with the existing division of academic labor that allocated psychology to philosophy, a subject to which he had made his own direct contributions.¹⁶ He saw his work in psychology as essentially another contribution to philosophy, although one that he hoped would have a major impact on that subject.¹⁷ What Wundt sought to achieve was a rejuvenation of philosophical inquiry by new means, not the constitution of a completely new discipline.¹⁸ For this the traditional object of psychological inquiry had to be preserved even as the means for pursuing that inquiry had to be radically changed.

A change in psychology's disciplinary project

Wundt's program for the reform of philosophy and the *Geisteswissenschaften* through the new psychology remained one man's idiosyncratic vision. It could please neither the more traditionally minded philosophers nor those modernists who considered philosophical goals irrelevant to the pursuit of knowledge through specialized scientific inquiry. In particular, Wundt's restrictions on introspection and experiment worked against the plausibility of his wider claims. These restrictions meant that the yield of his experimental program was actually rather limited. One could accept this with good grace and restrict the role of the experimental psychologist to that of a competent craftsman in a small number of specialized areas that lacked major intellectual or practical significance. Some, like G. E. Müller at Göttingen,¹⁹ adopted this course, but in the intensely competitive academic world of the nineteenth century this was not a generally useful prescription for either personal or disciplinary success. It could work on a large scale only where a discipline already possessed an established intellectual and institutional framework, but of course this was precisely what was lacking in the case of psychology.

Wundt's legacy to his students contained both positive and negative elements. On the positive side there were the institutionalized arrangements for conducting a psychological laboratory. But on a more strategic level Wundt's vision for the new psychology was of less help to his students

who faced a different world and had different ambitions. The root difference involved what is most appropriately called a “disciplinary project.” By this is meant the practitioners’ conception of how their discipline fits into the existing institutionalized structure of knowledge-generating domains and the nature of their discipline’s contribution to public knowledge and social practice.²⁰ Such a project may be an expression of already established patterns or it may be more on the level of the desirable, but in any case it provides the context that gives a broader meaning and direction to specific types of investigation.

Wundt’s disciplinary project for psychology did not provide for the severing of its links with philosophy and its development as an independent discipline. On the contrary, when some of the younger generation of experimental psychologists made proposals in this direction, Wundt opposed them vehemently.²¹ His methodological prescriptions were entirely consonant with his project for a psychology that was to retain intimate ties with philosophy and the *Geisteswissenschaften* – psychology was to be only partly an experimental science.

This project, however, proved to be out of step with the development of the social conditions on which it depended for its success. Wundt’s own appointment at Leipzig occurred at a time when the psychologizing of philosophy constituted a still-rising tide in Germany and aroused relatively little opposition.²² It is understandable that he then took these conditions for granted and based his project for the future of psychology on their continuation. That would have meant that a significant proportion of German university chairs in philosophy would gradually be occupied by psychologists. Such a process did in fact begin to get under way, but in due course it evoked a most determined reaction among the philosophical establishment who did not in the least share Wundt’s views on the relevance of scientific psychology for their own discipline.²³ His plans for the psychological rejuvenation of philosophy and the *Geisteswissenschaften* ran directly counter to the major internal developments in those fields, so that experimental psychology in Germany found itself rejected by the forces that were to have provided it with an institutional home base.²⁴ It was increasingly *forced* to make its own way as an independent discipline.

By contrast, in the United States, the advantages of independent disciplinary development were clear from the start, given a university structure that favored specialization and the absence (except on a local basis) of anything that could be compared with the entrenched German philosophical establishment. In Germany there had been a long period during which the sectarian religious control of university teaching had been replaced by the authority of philosophical systems, and in the late nineteenth century the institutional relics of this period were still strong. In American universities scientific – or, more properly, scientistic – dogma was apt to take over directly from religious dogma. Moreover, ultimate control of univer-

sity appointments and professional opportunities was vested in businessmen and their appointees, in politicians and in men engaged in practical professional activity, rather than in state officials whose vision had been shaped by their own philosophical education. Thus, the representatives of a new discipline like experimental psychology had to legitimize their project before a very different tribunal in the two cases.²⁵ Although the project of a scientific psychology emerged in both settings, it was much more clearly linked to the project of an *autonomous* discipline in America than in Germany.

Contrary to Wundt's conception, the new psychology did not prosper through the links with philosophy, linguistics, history, and anthropology that he had tried to forge. Instead, it shifted its weight to its other foot, as it were, and based its claims for recognition entirely on its affiliation with the natural sciences. In this it was greatly encouraged by a swelling tide of scientism during the closing years of the nineteenth century. The triumphal progress of the natural sciences helped to promote the belief that their methods were the *only* methods for securing useful and reliable knowledge about anything. Insights and explanations that could not be tied directly to these methods were empty speculations unworthy of serious attention. The split between the natural sciences and the social and humanistic disciplines had finally become an unbridgeable chasm and psychology was caught in the middle, forced to one side or the other. What Wundt had tried to keep together, however tenuously, now broke apart irrevocably. Unsurprisingly, most of those who had come to him primarily because of his laboratory, including virtually all his American students, ended up on the side of natural science and experiment, consigning the rest of Wundt's psychology to the graveyard of metaphysics.²⁶

There were two major themes in this surge of scientism. The one was technical-utilitarian, involving an assimilation of conceptions of truth to conceptions of usefulness and a consequent emphasis on practically applicable knowledge. Scientific knowledge was basically useful knowledge and thus superior to idle speculation. The second theme derived the superiority of scientific knowledge more from its close grounding in directly observed facts and its avoidance of airy metaphysical fancies that lacked this grounding.²⁷ Both themes were of course normative in the sense that they did not simply register what science was but prescribed what good or real science ought to be. It ought to be useful, and it ought to avoid theories that went beyond a summary of what had been observed. Both tendencies had their bogeyman. For the first, he was the ivory-tower intellectual; for the second, he was the philosopher spinning his metaphysical web.

These themes were obviously quite consonant with each other and they converged in their effect. But they appealed to different individuals in different degrees, depending on their backgrounds and life circumstances. Thus we find some of the first generation of post-Wundtian psychologists

putting more emphasis on one or other of these aspects. There were the practicalists, like G. Stanley Hall, who were often far from meticulous in gathering observations and not going beyond them, and there were the experimentalists, like Titchener, who were sticklers for laboratory finesse, but who cared little for any practical applications of psychological knowledge. Nevertheless, both sides were united in their rejection of the Wundtian synthesis. Psychology was to be a specialized positive science, divorced from philosophy, but looking to the natural sciences for its theoretical and methodological models.

Leaving the practicalist orientation for later consideration, we must briefly note the impact of the experimentalist orientation on investigative practice. For neither practicalism nor experimentalism could help changing the practice of Wundtian experimental psychology. As we have seen, the latter's practice had been based upon a strict limitation on the proper topics for experimental psychological investigation. The rest of psychology was left to be elucidated by *Völkerpsychologie* and by theoretical excursions into the mode of operation of "psychic causality." Now, both of these aspects were rejected as "unscientific" or "metaphysical," and experimental observation was to constitute the whole of psychology. But the traditional mentalistic objects of psychological investigation were not given up. The only way of reconciling these investigatory objects with the new emphasis on experimental exclusiveness was by greatly extending the scope of experimental introspection.

Dead end: Systematic experimental introspection

Some of Wundt's most influential students, notably Külpe in Germany and Titchener in America, took this path and thereby inaugurated a rather odd episode in the history of psychology's investigative practice, the period of "systematic experimental introspection." It was a brief period, for the methods that characterized it were at the forefront of disciplinary advance for at most a decade, from about 1903 to about 1913. But it was a fateful period because it revealed many of the problems inherent in the Wundtian synthesis of rigorous experimentation and mentalistic objects. The ensuing crisis virtually eliminated the Wundtian program and favored the development of psychology's investigative practices along altogether different lines.

One problem that we face in trying to understand the new approach to introspection on the part of some of Wundt's most prominent students is historiographic in nature. The behaviorist turn of American psychology produced a certain foreshortening of perspective on everything that preceded it. Like all effective leaders of movements, J. B. Watson made good use of a simplified image of the opposition. In due course it became customary for the most diverse positions to be lumped together under the

poorly understood category of "introspectionism." This had the effect of obscuring the very fundamental differences that existed between Wundt and his students, differences that were critical for the development of psychology as an independent discipline.

The change in the deployment of introspective methods certainly seemed very significant to those who helped to bring about the change. Here is how Titchener describes the situation in 1912:

Those who remember the psychological laboratories of twenty years ago can hardly escape an occasional shock of contrast which, for the moment, throws into vivid relief the difference between the old order and the new. The experimenter of the early nineties trusted, first of all, in his instruments; chronoscope and kymograph and tachistoscope were – it is hardly an exaggeration to say – of more importance than the observer. . . . There were still vast reaches of mental life which experiment had not touched; . . . meanwhile, certain chapters of psychology were written rather in the light of "system" than by the aid of fact. Now twenty years after we have changed all that. The movement towards qualitative analysis has culminated in what is called, with a certain redundancy of expression, the method of "systematic experimental introspection." . . . A great change has taken place, intensively and extensively, in the conduct of the introspective method.²⁸

What Titchener referred to as "extensive" changes included the extension of "systematic" introspective analysis to such areas as memory, thinking, and complex feelings, mostly topics that Wundt had explicitly designated as being unsuitable for the practice of experimental introspection. The Würzburg experiments were the most promising examples of this new fashion of intensive introspective analysis, but the vogue of qualitative introspection had many proponents and defenders who were not members of the Würzburg school.²⁹ In his relatively well-known attack on the methods of the Würzburgers, Wundt was of course expressing his opposition to the entire trend.³⁰ He reiterated that only when "the objects of introspection are directly tied to external physical objects or processes" did one have ideal conditions for experimental psychological investigation; all other circumstances compromise this ideal and need to be treated with great circumspection.

Among the "intensive" changes in the application of introspection that Titchener noted, two were particularly striking.³¹ One involved a shift in the relative importance attached to objective and to introspective data within a particular series of experimental observations. In the more traditional kinds of experiments the essential data had been either completely objective, such as reaction times and errors of recall, or tied directly to measured variations in physical conditions, as in psychophysics and various

experiments in perception. In terms of truly subjective data, the subject might at most be expected to give occasional reports on mental processes that accompanied his overt recorded responses. Systematic introspection changed this emphasis. Subjective reports were now required on a regular basis, usually for every experimental trial, and it was they, rather than more objective measures, that provided the essential data of the investigation. The studies of the Würzburg school, and those to which they gave rise,³² provide the best-known illustration of this trend, although it is not limited to this group.³³ In the Würzburg studies the actual solutions of the experimental tasks become almost irrelevant to the real purpose of the experiment, which is to provide subjective data on the process of thought. To Wundt, of course, these were simply "pseudoeperiments." The more cautious Müller warned against the danger of these methods coming to be considered the norm in experimental psychology.³⁴

Closely connected with the shift toward subjective reports is the second feature of the new introspectionism, the interest in qualitative description. Galton's studies of imagery³⁵ had to some extent anticipated this trend, but in the era of Fechner and Wundt they constituted a distinctly marginal area of psychological research. It was Binet who played a key role in challenging Wundtian introspection on its home ground. In 1903 he published a series of papers on investigations of the two-point threshold in which he had not limited himself to the customary introspective reports that went no further than noting whether one or two points had been felt upon application of the stimulus.³⁶ By questioning and through spontaneous reports from his subjects he accumulated a mass of qualitative data that showed that the conventional determination of the threshold represented a gross oversimplification of the subjective processes involved.³⁷ Binet's use of complex qualitative introspective material was quite similar in style to the methods of the Würzburg school, leading him to claim priority for himself.

The impulse that caused a significant number of experimental psychologists to turn toward "systematic" introspection at this time was the very opposite of "metaphysical." It was precisely because they wanted to replace the old-style speculation about matters like "psychic causality" by direct observation of mental events that they turned to qualitative introspective data. In doing this they were doing no more than following the time-honored basis of scientific experimentation, which provided for the creation of a distinct province of witnessed facts on which general consensus could be achieved.³⁸ Such a body of consensually validated facts would provide a much more satisfactory way of grounding disciplinary knowledge than this or that theoretical system. As is well known, the program of systematic introspection came to grief within just a few years. The reasons for this debacle are not without relevance for the history of psychology's investigative practices.

In the first place, we must distinguish between two kinds of criticism to which the practice of qualitative introspection was subjected. The first was primarily directed at the *object* of introspective practice, and the methodological aspect just followed as a consequence. Behaviorist criticisms of "introspection" were generally of this type. The fundamental objection was to making consciousness an object of scientific study. For J. B. Watson and most of his followers this rejection of psychology's traditional knowledge object was closely connected to their conviction that psychology had to be a practical science if it was to prosper as a discipline, and nothing practically useful had emerged or was likely to emerge from the analysis of consciousness.³⁹ The rejection of introspection as a method then followed as a matter of course, for the object on which it was to be practiced would cease to exist as an object for psychology. On the methodological side the behaviorist critique added nothing to what was already well known in the nineteenth century; the innovative aspect of this critique depended entirely on the radical change in psychology's investigative object.

In Europe there was little interest in a change of psychology's traditional knowledge object along behaviorist lines, but in many quarters there was dissatisfaction with the turn that experimental introspection had taken. This led to various kinds of criticism, some, like Wundt's, arguing for a return to the simpler certainties of a more traditional practice, others containing some new insights. These discussions did focus on the purely methodological aspects of introspection, sometimes entailing a change of investigative object as a consequence. In the present context we need consider only one line of criticism, which has a bearing on introspection as an investigative practice rather than merely as a technique.

From this point of view the critical development occurred when the problem of experimental introspection was recognized as essentially a problem of *communication* rather than a problem within the individual consciousness. The source of the difficulty did not lie in the privacy of the experiences that were observed but in the means available to establish a consensus that would transcend this privacy. In ordinary scientific observation the experiences that are described are private too – no two scientists see exactly the same phenomenon – but this problem is overcome by the use of special media of communication that make agreement on crucial aspects of the experience possible. These media are partly physical – relatively simple and artificially structured spatial patterns, for example – and partly linguistic. The role of the physical aspect had been recognized by Wundt and the more traditional experimentalists; now interest began to shift to the role of language. Was it possible to develop anything like a scientific language for describing the experiences evoked by "systematic" introspection?

The effective use of such a language would entail a certain attitude on the part of both participants in the experimental situation, experimenter

and subject. The desired description would be of the kind that is used in talking about an object toward which one maintains a certain distance. Now, for the experimenter there is no special difficulty in talking about the subject's experience in this way. But for the subject there is, because it is his own experience from which he can only distance himself by adopting a special attitude, the introspective attitude. This is certainly possible, but what are the costs? For costs there must be, because something is lost between the original experience and the report to the experimenter. In the observational reports of natural science, much of the quality of the observer's experience is sacrificed too in the interests of reaching agreement on the essentials. But for traditional psychology the qualities of the original experience *were* the essentials. The achievement of the kind of consensus on the "facts" that science demanded would have entailed the abandonment of the object one was supposed to be investigating.

Proponents of "systematic experimental introspection" faced a dilemma. They could either insist on the use of the kind of distanced language that had been useful in achieving consensus in the natural sciences, or they could reject such limitations and try to remain close to the phenomena as experienced. In the first case they lost the real object of their scientific interest; in the second case they lost the possibility of achieving a scientific consensus on the "facts."

In practice the language of scientific description turned out to be the language of sensationism. By reducing lived experiences to reports on specific sensational components, systematic introspection could imitate the path taken by the observational reports of natural science. This policy was essentially the one advocated by Titchener and put into practice at his Cornell laboratory.⁴⁰ But Titchener was only able to argue in this way because he had reified the terms of his sensationist language and persuaded himself that the basis of conscious experience really was nothing more than an assembly of sensations. He now faced a war on two fronts, which could only end in total defeat. His fellow introspectionists refused to sacrifice their phenomenal objects at the altar of sensationist reductionism, and his more practically minded colleagues refused to be interested in a constructed object of investigation that had no relevance to everyday life.

"Systematic experimental introspection" was based on an attempt to extend the essentially public project of scientific witnessing to the most private areas of experience – complex and indefinite thoughts and feelings. But that would mean providing the same kind of unambiguous, analytical description of such experiences as one would give of a simple visual stimulus in a perception experiment. It was soon realized that if one did not, like Titchener, insist on such descriptions, the spontaneous reports of the subjects in these experiments were of a very different order. They were not at all like descriptions of an object regarded from a distance but more like a verbal expression that somehow belonged to the experience itself. Asked

to report on their thoughts on receiving the experimental task, subjects would say things like: "Wait a minute, there's a trap here," or "well, really!" or "OK, here's another one of those Nietzschean paradoxes." But as von Aster pointed out in a brilliant paper,⁴¹ the relation of the verbal "report" to the experience is quite different here than in the case where the subject reports, let us say, on the size of a presented rectangle or the length of a line. In the first examples the subject is giving *expression* to the Gestalt quality of his experience; in the second case he is *describing* specific elements of his experience. Note that he uses a metaphorical, expressive kind of language in the first case. He is not really describing something at a distance, he is giving public expression to a lived experience.

This insight led to the important distinction between *Beschreibung* ("description") and *Kundgabe*. The second term has no precise equivalent in English. It combines the connotations of "expression" and "communication." It refers to a process of giving public expression, a process of declaration rather than reporting. Its language, as von Aster noted, must necessarily be metaphorical and hence ambiguous, because we have no other way of giving verbal expression to an experience as a lived whole. But these features seem to render the language of *Kundgabe* unsuitable for the purposes of scientific communication. Only the neutral, analytic language of distanced description would do for that. But this language destroys what was supposed to be the object of investigation.⁴²

"Systematic experimental introspection" therefore faced an impasse. If one rejected the Titchenerian destruction of actual lived experience, however, there were two possibilities that remained. One could try to develop a descriptive language that, while reasonably unambiguous, allowed the complex, wholistic qualities of experience to enter public scientific discourse. In due course, the Gestalt psychologists showed that this could be done, although it was done most successfully where the link to physical stimuli existed, in the study of external perception. In the area of thinking, it was always more difficult to carry through this program convincingly and consistently.

The second possibility opened up by a critical insight into the problems of "systematic experimental introspection" depended on an acceptance of the nature of *Kundgabe*. If it was true that in many psychologically interesting situations the subject's verbal statements were not really *about* his or her experience so much as an *expression of* that experience, then those verbal statements themselves should become the objects of psychological study. But they would be studied for what they were, a manifestation of a living personality, rather than as a source of information about a separate world of private consciousness. This of course would mean the abandonment by psychology of its traditional, investigative object, the world revealed by the "inner sense." In Europe this path led to Kurt Lewin's studies of motivated behavior as well as to some versions of clinical existential-

ist psychology. In America the whole development was of course short-circuited by the behaviorist turn. But one did not need to be a behaviorist to abandon psychology's traditional object of study.

These developments entailed a shift in the structure of the psychological experiment as a social situation. In the Wundtian experiment, the experimental subject was the scientific observer, and the experimenter was really a kind of experimental assistant. Only the person reacting to presentations of the stimulus material could observe the phenomena of interest and report on them. But if we start treating the experimental subject's report as *Kundgabe*, as an expression of rather than a descriptive report about his experience, then he is no longer in the role of a scientific observer. That role has now passed to the experimenter who records and then reports on what the subject said. It was noted at the time that this was happening in the Würzburg experiments.⁴³ The implications of this change were extremely far reaching, for it demonstrated that the nature of the object of psychological investigation was linked to the social structure of the investigative situation. A fundamental change in the one entailed a fundamental change in the other. As the object of psychological investigation changed, so the distribution of tasks in the investigative situation would have to change. The social structure of the Wundtian experiment could not survive the abandonment of its investigative object.

4

The social structure of psychological experimentation

How the Leipzig research community organized itself

In the last section of chapter 2 we took note of the fact that the establishment of psychological laboratories involved the institutionalization of certain social arrangements. Psychological experimentation became a collaborative effort dependent on a division of labor among individuals who carried out different functions in the experimental situation. More specifically, a broad distinction emerged between those who acted as the human source of psychological data, the experimental subjects, and those who manipulated the experimental conditions, the actual experimenters. Although this distinction was essentially a response to the practical exigencies of late nineteenth-century “brass instruments” experimentation and not the product of deep reflection, it rapidly became traditionalized. To carry out sophisticated experiments with relatively complex apparatus, one now had to assign the function of data source and the function of experimental manipulation to different people.

What was not noticed at the time, or indeed for almost a century afterward, was that this arrangement created a special kind of social system – one of psychological experimentation. The interaction between subjects and experimenters was regulated by a system of social constraints that set strict limits to what passed between them. Their communication in the experimental situation was governed by the roles they had assumed and was hedged around by taken-for-granted prescriptions and proscriptions.

However, the specific features of this social system were not necessarily fixed. The basic division of labor between experimenters and subjects still left much room for local variation. For instance, there was nothing in the practical requirements of psychological experimentation that dictated a *permanent* separation of experimenter and subject roles. The practical